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Insight

Hybrid Bonding Expands from Logic to Memory: SK Hynix, Applied Materials, BESI Drive Co-optimization to Scale Next-gen HBM

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Summary

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- *Memory makers are actively pursuing hybrid bonding tools as a key differentiator, with SK Hynix signaling the shift toward production-scale adoption.*
- *SK Hynix's tool procurement signals a shift toward scaling hybrid bonding for high-volume HBM manufacturing.*
- *Integrated solutions from Applied Materials and BE Semiconductor Industries to reduce process complexity and address key CMP-driven challenges.*

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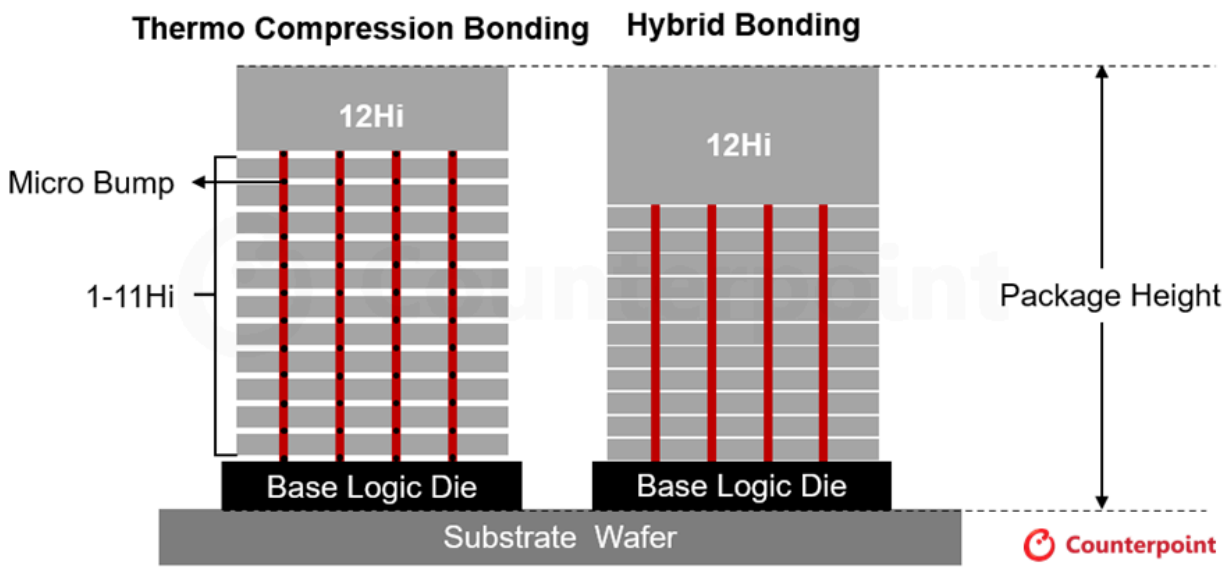
High Bandwidth Memory (HBM) has become a critical enabler for AI and high-performance computing (HPC), delivering the bandwidth and efficiency required for data-intensive workloads like large language models. HBM typically stacks 12-16 DRAM dies on a base logic die and is integrated alongside GPUs. Currently, these stacks rely on micro-bump interconnects using methods such as mass reflow with moulded underfill (MR-MUF) or thermal compression bonding with non-conductive film (TC-NCF).

However, micro-bumps are emerging as a key bottleneck limiting signal integrity, power efficiency, and thermal performance. As stack heights increase to 20+ layers, physical height and integration constraints within GPU packages are becoming more pronounced.

Thermo Compression Bonding vs Hybrid Bonding



Sources: Counterpoint Research, BESI

Shift Toward Hybrid Bonding:

To overcome these limitations, the industry is moving toward hybrid bonding, which enables:

- Finer-pitch interconnects
- Reduced die-to-die spacing
- Lower stack height
- Higher bandwidth and energy efficiency

While JEDEC has relaxed HBM height specifications allowing the continued use of TCB for up to 16 layers leading memory players like Samsung Electronics, SK Hynix, and Micron Technology are actively advancing hybrid bonding for HBM4 and beyond to meet next-generation AI demands.

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In hybrid bonding, CMP becomes mission-critical, as bonding relies on ultra-flat, atomically smooth surfaces.

Any CMP defect like Cu dishing, erosion, or contamination directly impacts yield, resistance, and reliability. This elevates CMP from a process step to a determinant of performance, interconnect density, and scaling in HBM4 and beyond.

Co-optimization: Applied Materials + BE Semiconductor Industries (BESI)

The collaboration between Applied Materials and BESI integrates CMP, surface preparation, and hybrid bonding into a single, controlled flow.

- Combines Applied Materials' strength in CMP and materials engineering with BESI's leadership in hybrid bonding and die placement
- Reduces process complexity, variability, and defect risks
- Improves alignment accuracy and bonding yield

The procurement of integrated hybrid bonding tools by SK Hynix reflects a strategic move to address key hybrid bonding challenges, particularly around CMP-driven surface uniformity, Cu dishing, particle defects, and ultra-tight roughness control. The integrated platform directly tends to tackle these critical bottlenecks, enabling more stable manufacturing, reducing yield risk, and accelerating the transition of hybrid bonding into high-volume production.

Key Challenges:

- HBM standard relaxation risk: Joint Electron Device Engineering Council (JEDEC's) increase in allowable HBM height could delay hybrid bonding adoption, extending TCB's lifecycle and creating uncertainty in equipment demand.
- Process sensitivity: Hybrid bonding is highly sensitive to CMP quality defects like dishing, contamination, and voids which increase resistances and impact yield despite superior electrical performance.

Impact on Memory Players:

In the near term, delaying hybrid bonding can:

- Reduce capital intensity
- Leverage mature TCB processes for better yield

However, long-term adoption remains inevitable due to enhanced performance requirements.

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the current phase of the AI infrastructure boom. According to **Counterpoint Research Report on Memory Solutions for Gen AI**, SK Hynix continues to lead in high bandwidth memory, crucial for generative AI with innovations including front-end process improvements in memory cell design, back-end stacking enhancements and integration of logic circuits (IVC) in base dies for voltage control. Further, the memory squeeze is intensifying as AI deployment accelerates, leading to a structural supply shortage lasting through 2030. However, 2028 will likely serve as the inflection point, as a new wave of production clusters begins to ramp up.

HBM5 Will be the True Inflection Point for Hybrid Bonding:

HBM5 is going to be the true inflection point for hybrid bonding, enabling >20-layer stacking, finer pitch, and next-level bandwidth while requiring breakthroughs in CMP, yield, and cost to scale into high-volume manufacturing.

For SK Hynix, early adoption of integrated solutions from Applied Materials and BESI provides a strategic advantage enabling it to meet future requirements in bandwidth, latency, power, and speed, regardless of interim standard relaxations. Leveraging Applied's process modules (CVD, PVD, ECD, CMP, surface prep) expertise along with Besi's hybrid bonder technology, will help build an essential foundation for achieving the stability and consistency required in advanced, high-volume chip production.

We expect SK Hynix to introduce HBM5 in 2029-2030, aligned with next-generation AI GPU cycles, with hybrid bonding reaching mass adoption at the HBM5 node to enable higher performance and efficiency.

Conclusion:

HBM scaling beyond ~16 layers and rising AI performance demands are pushing micro-bump packaging to its limits. Hybrid bonding offers a scalable path forward with:

- Sub-10μm interconnect pitch (vs 20μm-40μm for micro-bumps)
- Higher bandwidth and power efficiency
- Compatibility with logic-centric integration

As HBM moves closer to logic-node requirements, hybrid bonding becomes essential to sustain performance scaling. The co-optimization of CMP and bonding led by Applied Materials and BESI is emerging as the key enabler of this transition.

SK Hynix's procurement of integrated hybrid bonding tools from Applied Materials and BE Semiconductor Industries will address critical challenges in HBM manufacturing while providing a strategic edge to meet next-generation performance needs and sustain HBM leadership, driving revenue growth and market share gains.

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Ashwath Rao has more than 20 years working experience in industry, research and academics. Ashwath had an opportunity to work with Intel Technology. Ashwath's major coverage in Counterpoint is semiconductors...

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